



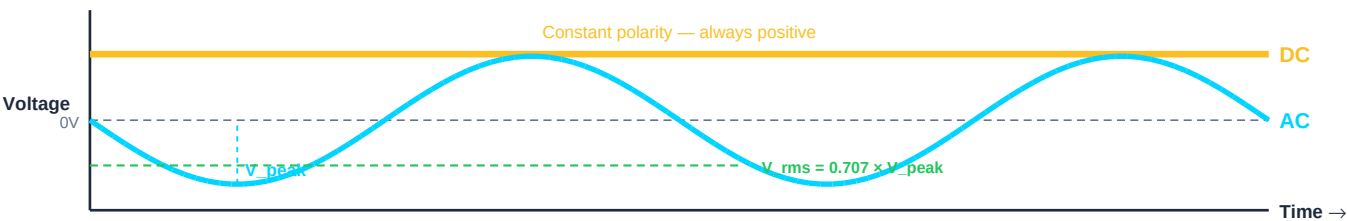
AC vs DC

Transformers · 3-Phase · Power Factor · Reactive Power

• ELECTRICAL ENGINEERING • LESSON 04

AC vs DC — The Fundamental Difference

Direct Current (DC) flows in one direction only. The voltage stays constant (or varies slowly). Batteries, solar panels, and electronic circuits run on DC. **Alternating Current (AC)** reverses direction periodically in a sinusoidal pattern. The mains supply in every building worldwide is AC — either 50 Hz or 60 Hz.



Property	DC	AC
Direction of flow	One direction (+ to −)	Reverses at supply frequency (50/60 Hz)
Voltage	Constant (e.g. 12V, 24V, 48V)	Sinusoidal: described by peak and RMS values
Transmission loss	High — voltage cannot easily be stepped up	Low — transformers step voltage up/down efficiently
Generation	Batteries, solar PV, rectifiers	Generators, alternators, inverters
Standard supply	12–48 V (vehicles, telecoms, solar) >100 kV (HVDC long-distance)	230 V / 50 Hz (UK, EU, Africa) 120 V / 60 Hz (USA, Canada)
Used for	Electronics, EVs, solar storage, LED drivers	Building supply, motors, heating, lighting
Transformer works?	No — transformers need changing flux	Yes — core reason AC dominates grid distribution

Why AC Won the "War of Currents"

In the 1880s, Edison backed DC and Tesla/Westinghouse backed AC. AC won for one decisive reason: **transformers**. A transformer can step AC voltage up to 400 kV for long-distance transmission (reducing I²R losses by 1 000×), then step it back down to 230 V for homes — all with >98% efficiency and no moving parts. No equivalent DC solution existed until modern power electronics.

Grid Voltage Level	Typical Voltage	Purpose
Generation	11–25 kV	Power station output (generator terminal voltage)
Transmission	132–400 kV	Long-distance bulk power transport (minimise I²R loss)
Sub-transmission	33–132 kV	Regional distribution to large industrial sites
Primary distribution	11–33 kV	Urban and suburban distribution networks
Secondary distribution	400/230 V	Final step-down to consumers via pole/pad-mount transformer
HVDC link	±320–±600 kV	Modern long sub-sea or intercontinental DC links (>99% eff.)

Transformer turns ratio: $V_1/V_2 = N_1/N_2 = I_2/I_1$. A transformer that steps voltage up by 10× steps current down by 10×. Power is conserved: $P_{in} = P_{out}$ (ignoring small losses). This is why power is transmitted at high voltage — lower current means lower I²R heating in cables.

Three-Phase AC — Why It Powers Industry

Single-phase AC (one live + neutral) is used for homes and light commercial loads. **Three-phase AC** (three lives, 120° apart) is used for all motors above ~2 kW, industrial equipment, data centres, and grid distribution. It delivers more power with less conductor material and produces a smooth, constant torque in motors.

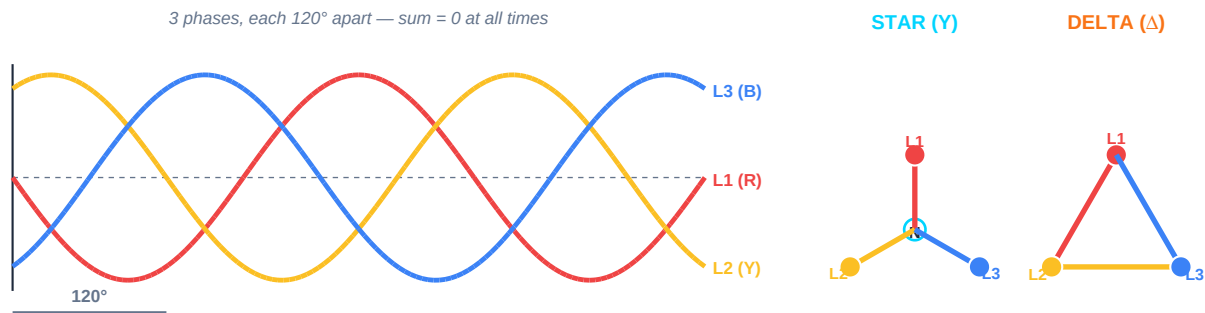


Figure 2 — Three-Phase AC Waveforms and Star/Delta Connections

Property	Single-Phase	Three-Phase
Conductors needed	2 (live + neutral)	3 or 4 (3 lives + optional neutral)
Power formula	$P = V \times I \times \cos \phi$	$P = \sqrt{3} \times V_L \times I_L \times \cos \phi$ (balanced load)
Voltage (UK/EU)	230 V line-to-neutral	400 V line-to-line / 230 V line-to-neutral
Motor behaviour	Pulsating torque; needs capacitor or split-phase start	Smooth, self-starting torque; no capacitor needed
Conductor material	Baseline = 1.0	Delivers same power with ~75% of conductor weight
Star (Y) connection	N/A	Provides both 400V (line-line) and 230V (line-neutral) from same supply
Delta (Δ) connection	N/A	400V line-to-line only; no neutral; balanced loads only

Transformer Fundamentals

Parameter	Formula	Example (11 kV → 400 V distribution transformer)
Turns ratio	$a = N1/N2 = V1/V2$	$a = 11\,000/400 = 27.5$
Secondary current	$I2 = I1 \times (N1/N2)$	If $I1=10A$: $I2=10 \times 27.5=275A$ (power stays same)
kVA rating	$S = V1 \times I1 = V2 \times I2$	$11kV \times 10A = 110kVA$; $400V \times 275A = 110kVA$
Efficiency	$\eta = P_{out}/P_{in} \times 100\%$	Modern distribution transformer: 98.5–99.5%
Voltage regulation	$VR\% = (V_{NL} - V_{FL})/V_{FL} \times 100$	Good transformer: $VR < 4\%$; poor: $> 10\%$

Single-phase vs three-phase connection: In a UK installation the supply enters at 400 V three-phase. Individual domestic properties are connected to one phase (230 V). Industrial and commercial premises are connected to all three phases to allow 400 V motors and balanced loading across the grid.

Power Factor — The Hidden Energy Thief

In a purely resistive circuit (heaters, incandescent lamps), voltage and current are in phase and all power is usefully converted. In circuits with inductance (motors, transformers) or capacitance, current is phase-shifted relative to voltage. The portion of power that does useful work is called **active power (P)**; the wasted oscillating energy is **reactive power (Q)**.

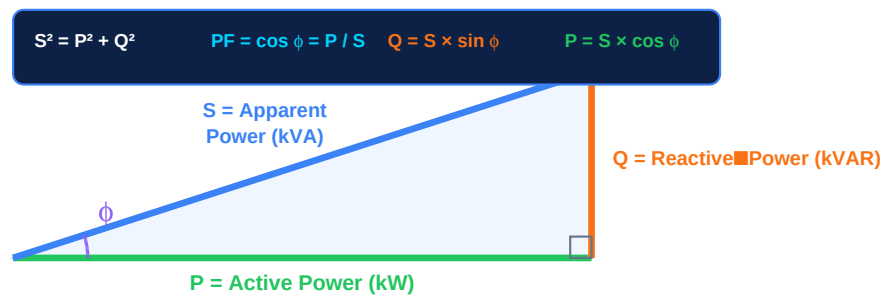


Figure 3 — Power Triangle: Active P, Reactive Q, Apparent S and Power Factor

Power Type	Symbol	Unit	Definition	Measured By
Active Power	P	kW	Real work done: heat, light, mechanical motion. $P = S \times \cos \phi$	kWh meter (your electricity bill)
Reactive Power	Q	kVAR	Energy oscillating between source and magnetic/electric fields. $Q = S \times \sin \phi$	kVARh meter (utility charges large users)
Apparent Power	S	kVA	Total power drawn from supply. $S = V \times I = \sqrt{(P^2 + Q^2)}$	Generator and transformer ratings
Power Factor	PF	(0–1)	$PF = P/S = \cos \phi$. Unity (1.0) = perfect. Low PF = poor efficiency.	Power analyser / smart meter

Power Factor Correction

A low power factor means the supply cable and switchgear carry more current than necessary to deliver the same useful power — causing higher cable losses and requiring larger (more expensive) equipment. Utilities charge large industrial customers a penalty for low PF. **Power factor correction (PFC)** adds capacitors in parallel with inductive loads to cancel reactive current.

PF Value	Load Characteristic	Consequence	Correction Method
1.0 (unity)	Pure resistive (heaters, incandescent lamps)	Ideal — no reactive current penalty	No correction needed
0.85–0.95	Good industrial load (efficient motors, LED with PFC driver)	Acceptable — typical utility target; small penalty possible	Minor PFC capacitor bank
0.70–0.85	Moderate inductive load (older motors, fluorescent without PFC)	Noticeable cable overloading; utility may charge penalty tariff	Fixed or automatic capacitor bank
< 0.70	Heavy inductive load (arc furnaces, large transformers on light load)	Significant over-sizing of all equipment; high utility penalty	Large automatic PFC bank or active filter (STATCOM)

PFC Calculation	Formula	Example
Required capacitor kVAR	$Q_c = P \times (\tan \phi_1 - \tan \phi_2)$	$P=100\text{kW}$, improve PF 0.75→0.95: $\phi_1=41.4^\circ$, $\phi_2=18.2^\circ$ $Q_c=100 \times (0.882 - 0.329) = 55.3$ kVAR

Capacitance (single-phase)	$C = Q_c / (2\pi \times f \times V^2)$	55.3kVAR at 230V/50Hz: $C = 55300 / (2\pi \times 50 \times 230^2) = 3.32 \text{ mF}$
Capacitance (three-phase star)	$C = Q_c / (3 \times 2\pi \times f \times V_{\text{phase}}^2)$	Same Q_c at 400V (star): $C = 55300 / (3 \times 314 \times 230^2) = 1.11 \text{ mF per phase}$

What comes next — ELEC-005: Electrical Safety — electric shock hazards, earth/ground systems, RCDs, fuses vs MCBs, PAT testing, and the IEC 60364 / BS 7671 wiring regulations framework.